PO GOVERNANCE × SAP · DEVELOPER BRIEF

Stage 1 (BW), Stage 2 (SAP R/3) and Stage 3 (AI): what to build

A plain, direct specification for the developer. It says what data comes in, from where, in which format, what the server does with it, and what it sends back to SAP. The full field-level API is in the companion file **PO_Governance_SAP_API_Specification_v1.4.pdf** and **po-governance-sap-api.yaml**.

Audience: developer, SAP BW / ABAP / workflow / Basis teams 9 Oct 2026

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01 What the tool does

The tool checks every PO before it is approved. It answers one question: **will this quantity push stock cover for this article-colour too high?** It gives a verdict (Allow, Hold, Watch List or Block) and, where possible, the maximum quantity that is still acceptable.

To do this the server needs three things:

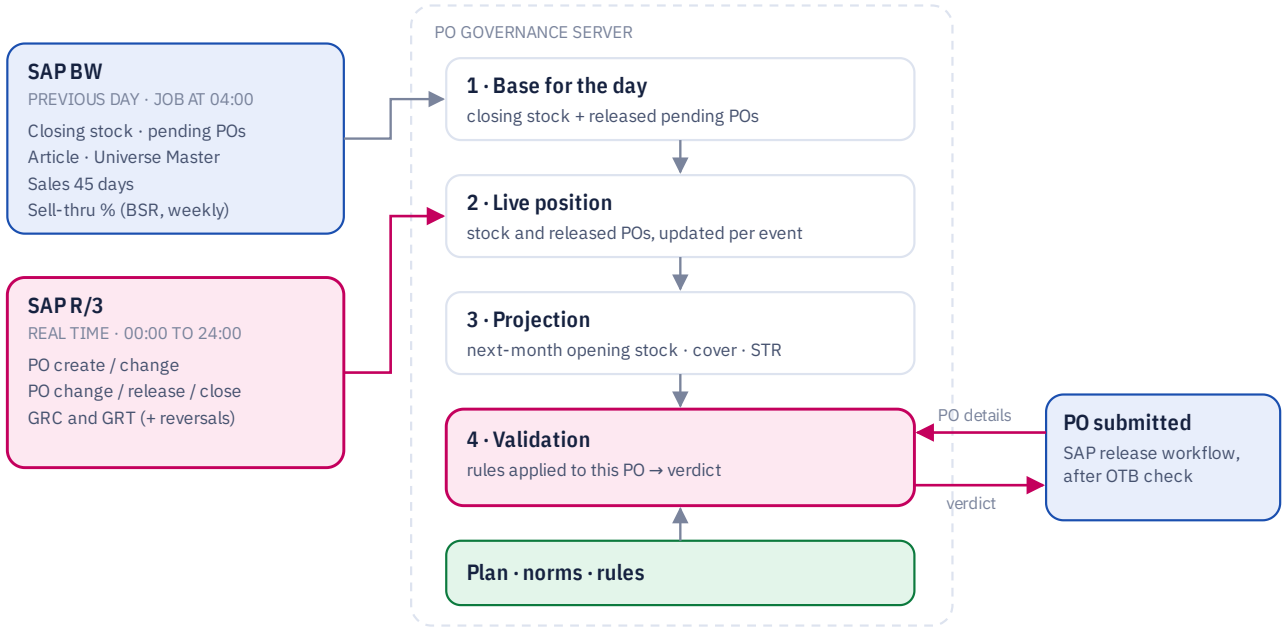
1. **Stage 1 · BW (daily at 04:00):** previous day's closing stock, released pending POs, Article Master, Universe Master, sales trend (last 45 days) and sell-thru % (BSR, weekly). This is the base for the day. Colour Master comes from SAP; the Plan from SAP SFCC / SFCR every hour (later; manual upload now).

2. **Stage 2 · SAP R/3, in real time:** every change to released POs, goods receipt (GRC) and goods return (GRT) during the day, plus a call from the PO release workflow asking for a verdict.
3. **Stage 3 · AI layer, later:** reads the decisions and exceptions already recorded, explains them and suggests rule changes. No new SAP interface.

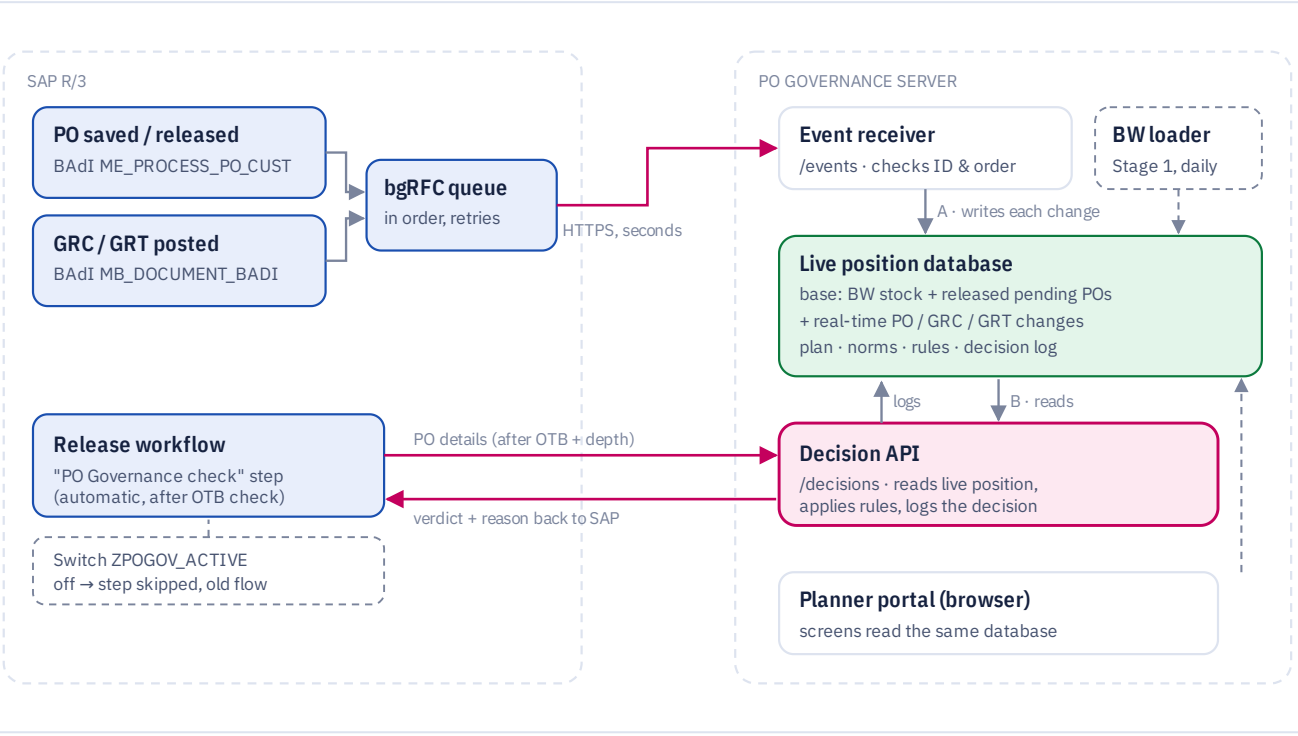
02 Fixed rules

RULE	WHAT IT MEANS FOR THE BUILD
OTB stays in SAP	SAP checks OTB first. The tool is called only after the OTB check passes. Never send OTB data to the tool.
Depth plan stays in SAP	SAP checks the minimum depth per option. The tool does not hold depth data. The depth check is bypassed only in the two cases in section 06
Open PO = released POs only	"Open PO" always means the **pending quantity of released POs** (ordered – received + returned, for POs not closed or deleted). A PO that is not yet released counts as zero
Company level	Stock and POs are added up across all DCs and stores, per article-colour. Store-level stock is not needed.
Vendor POs only	Only vendor POs and goods movements against vendor POs are sent. Stock transfers between DCs and stores (STO) are not sent.
PO, GRC and GRT together	A GRC moves quantity from open PO to stock. All three must come through the same real-time feed, in order.
No real-time sales	Sales for the last 45 days from BW are enough.
Plan	Uploaded manually now. Later fetched from SAP (SFCC / SFCR) **every hour**; it does not need real-time sync.
The tool can never stop SAP	If the tool is slow, down or unsure, SAP continues with the current process (section 11).
Decision rules	Exactly as in VMM_PO_Governance v89 (6). See the Functional Logic Guide.

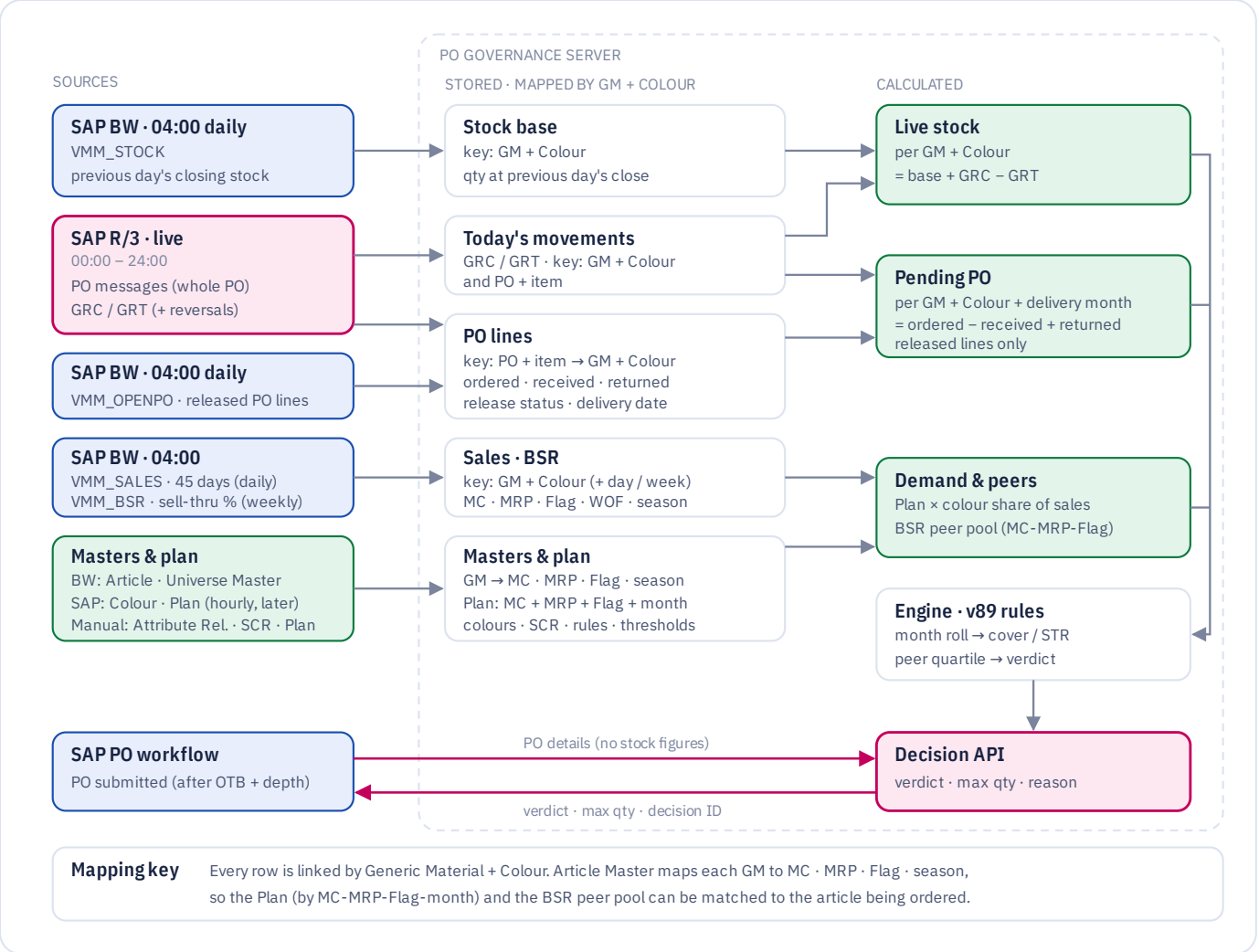
03 How it fits together



Inside the server, the event receiver and the Decision API share **one database** (the live position). The receiver writes to it; the Decision API reads from it. Nothing is copied between them.



How data is mapped on the server



1. **Receive:** BW files at 04:00 (checked against the control file), live R/3 messages all day, the plan (manual now, hourly from SAP later) and the manual uploads.
2. **Store:** each source goes into its own table. Stock, sales and BSR are keyed by **GM + Colour**. PO lines are keyed by **PO + item** and carry their GM + Colour. GRC / GRT are stored against both.
3. **Link:** Article Master (from BW) adds MC · MRP · Flag · season to every GM, so the Plan (by MC-MRP-Flag-month) and the BSR peer pool can be joined to the article.
4. **Calculate:** live stock, pending PO by delivery month, colour demand and the peer pool, per GM + Colour.
5. **Decide:** when SAP submits a PO, the v89 engine reads these figures, and the Decision API returns the verdict and max quantity.

04 Stage 1: data from BW and SAP masters

All data and where it comes from

#	DATA	SOURCE	HOW OFTEN	USED FOR
1	Closing stock	BW	Daily, 04:00 (previous day's close)	Base stock for the day
2	Pending POs (released)	BW	Daily, 04:00	Base open PO for the day
3	Article Master	BW	Daily, 04:00	Article details, attributes; links GM to MC · MRP · Flag · season
4	Universe Master update	BW	Daily, 04:00	Decides Single Concept vs Multi Concept
5	Sales trend (last 45 days)	BW	Daily, 04:00	Colour mix and sales trend
6	Sell-thru % (BSR)	BW	Weekly	Peer ranking (Multi Concept)
7	Plan (SFCC / SFCR)	Manual upload now; later SAP	Later: **every hour**	Future demand per MC-MRP-Flag-month
8	Colour Master	SAP (all SAP colours)	Daily, 04:00	Colour codes and names; checks every colour received
9	Attribute Relevance Master	Manual upload	When changed	Multi Concept recommendations
10	SCR mapping / category heads	Manual upload	When changed	Logins and approval routing in the tool
—	PO changes, GRC, GRT	SAP R/3 live	Real time, 00:00 – 24:00	Live position during the day (section 05)

How the files are delivered

- BW adds one step at the end of its process chain: an **Open Hub export to CSV**.
- **Timing:** the BW job runs at **04:00 every day** and extracts the **previous day's closing snapshot**. Example: at 04:00 on 9 Oct, BW extracts the 8 Oct closing stock and pending POs (all postings dated up to 8 Oct) and sales up to 8 Oct.
- Files go to a **shared folder on the tool server**, or by **SFTP**. The BSR file comes once a week (on a fixed day agreed with BW).
- The Colour Master is exported from SAP with the 04:00 files (from BW if available, else an R/3 extract).
- **Plan (later):** an hourly job exports the SFCC / SFCR plan from SAP to the same folder. The server loads it only when it has changed. Until then, planning uploads it manually.
- Format: CSV, UTF-8, header row, fixed column order, dates `YYYY-MM-DD`, no thousand separators.
- File names: `VMM_<DATASET>_<YYYYMMDD>.csv`, e.g. `VMM_STOCK_20261009.csv` (hourly plan: `VMM_PLAN_<YYYYMMDDHH>.csv`).
- A **control file** is written last: `VMM_CONTROL_<YYYYMMDD>.csv`. It lists only the files sent in that delivery.

The files

FILE	COLUMNS	FREQUENCY
VMM_STOCK	Generic Material, Article, Colour, Brand Type, Division, MC CD, MRP, Flag (BOGO / non-BOGO), Stock Qty, Stock Value, As-of Date. One row per article-colour (site rows allowed; the server adds them up)	Daily
VMM_OPENPO	PO Number, PO Item, Generic Material, Article, Colour, Brand Type, MC CD, MRP, Flag, Vendor, Ordered Qty, Received Qty, Returned Qty, Pending Qty, Delivery Date, Release Date. **Released, not closed, not deleted vendor PO lines only.** Received and Returned count only postings dated up to the snapshot day (8 Oct), so they match the stock file	Daily

FILE	COLUMNS	FREQUENCY
VMM_ARTICLE	Generic Material, Article, Colour, Brand Type, Division, MC CD, MC Desc, MRP, Flag, Season, Style, Fit, Product Material, Design, Vendor, Vendor Name	Daily
VMM_UNIVERSE	Generic Material, MC CD, MRP, Flag, Concept (Single), Valid From	Daily
VMM_SALES	Generic Material, Colour, Brand Type, MC CD, MRP, Flag, Date, Sales Qty. Last 45 days, one row per day	Daily
VMM_BSR	Generic Material, Colour, MC CD, MRP, Flag, Weeks on Floor, Avg Per-Day Sell-Thru %, Season, Snapshot Week	Weekly
VMM_COLOUR	Colour Code, Colour Description (all colours in SAP)	Daily
VMM_PLAN (later)	MC CD, MRP, Flag, Month, Plan Qty, Plan Version (SFCC / SFCR). The server splits the month into days using historical day mapping	Hourly
VMM_CONTROL	File Name, Row Count, Total Qty, **Snapshot Date** (the business day the data closes, e.g. 8 Oct), **Extraction Time** (when the job actually ran, e.g. 9 Oct 04:00)	With every delivery

What the server does with them

1. Wait for the control file. Do nothing until it exists.
2. Check every listed file's row count and total quantity against the control file.
3. All match: load them as the new base, in one database transaction. Save the snapshot date and extraction time. Drop the real-time adjustments of the snapshot day (8 Oct: they are inside the new snapshot). Keep today's (9 Oct) messages and re-apply them on the new base.
4. The BSR file is replaced only when a new weekly file arrives. Between weeks, the last BSR snapshot is used. The hourly plan file replaces the plan only if its content has changed.
5. Masters (Article, Universe, Colour) are loaded before stock and POs, so every row can be linked. A row with an unknown article or colour is held aside and reported, not dropped silently.

6 Any mismatch or missing file: keep the previous base, show the data date on screen, and email the BW team and planning. Manual upload still works.

05 Stage 2a: SAP real-time messages

Real-time messages

SAP sends a message to the server every time one of these documents is saved.

SAP EVENT	MESSAGE TYPE	WHERE TO HOOK IN SAP
PO created / changed	`PO_CREATED`, `PO_CHANGED`	BAdI `ME_PROCESS_PO_CUST`
PO released / release revoked	`PO_RELEASED`, `PO_RELEASE_REVOKED`	Release BAdI / workflow event
PO closed / deleted	`PO_CLOSED`, `PO_DELETED`	BAdI `ME_PROCESS_PO_CUST`
Goods receipt / reversal (101 / 102)	`GR_POSTED`, `GR_REVERSED`	BAdI `MB_DOCUMENT_BADI`
Return to vendor / reversal (122 / 161)	`RETURN_POSTED`, `RETURN_REVERSED`	BAdI `MB_DOCUMENT_BADI`
Verdict follow-up (buyer reduced qty, exception approved / rejected)	`GOV_OUTCOME`	Workflow step

How SAP sends them

- Write the message **after commit**, into one ordered **bgRFC queue**. The user never waits.
- The queue sends it to `POST /api/sap/v1/events` over HTTPS and **retries until the server answers 200**.
- Every message has a unique `msg_id` and an increasing `seq`. The server ignores a message it has already seen.
- A PO message carries the **whole PO** (header and all items), not just what changed.

Key fields

FIELD	MEANING
`msg_id, seq, event_type, occurred_at, sap_user`	Message identity, order, type, time saved in SAP, user
`purchase_order.po_number, vendor, purch_group, release_status`	PO header (EKKO). The server uses release_status to decide whether the PO counts
`items[].po_item, article, generic_article, colour, merch_category, site`	What is ordered (EKPO). Site is informational
`items[].order_qty, received_qty, returned_qty, open_qty, delivery_date`	Quantities and first delivery date (EKPO / EKBE / EKET)
`items[].deletion_flag, delivery_completed`	Flags (LOEKZ / ELIKZ)
`goods_movement.material_doc, movement_type, quantity, debit_credit, posting_date, po_number, po_item, reversal_of`	One material-document line (MSEG / MATDOC)

What the server does with each message

MESSAGE	EFFECT
PO released	The PO's pending qty is added to open PO for its article-colour and delivery month
Released PO changed (qty or date)	Overwrite that PO's lines with the new state; open PO is recalculated
PO created / changed, not yet released	Stored for reference only. **No effect** on open PO
Release revoked / PO closed / deleted	That PO's pending qty is removed from open PO
GRC (101) / reversal (102)	Live stock + qty / − qty. The PO's pending qty − qty / + qty

MESSAGE	EFFECT
GRT (122 / 161) / reversal	Live stock – qty / + qty. Pending qty + qty / – qty if the PO is still released and open

- The live feed runs **24 hours, 00:00 to 24:00**. Every message is stored with its business day, and applied by the rule in section 07 ("No double counting").
- Apply each message in **one database transaction**. Answer 200 only after the commit.
- A message that fails validation is **parked** (stored, reported to IT). It never blocks the queue.

06 Stage 2b: check when a PO is submitted

The PO release workflow gets one new automatic step, **"PO Governance check"**, placed **after** SAP's OTB check and depth plan check.

Request: `POST /api/sap/v1/decisions`

FIELD	MEANING
`request_id` *	Unique per check, e.g. work item + PO + version. Same id + same body returns the same answer
`mode` *	`SHADOW` (decide and record only) or `LIVE` (SAP routes on the answer)
`otb_check` *	Always `PASSED` (the tool is called only after OTB passes)
`depth_check` *	`PASSED`, or `BYPASSED` when a tool-approved quantity is below the minimum depth
`previous_decision_id`	The earlier decision, when the PO is resubmitted after a reduction
`purchase_order: po_number, buyer_user, division, vendor; items[]: po_item, article, generic_article, colour, merch_category, season, order_qty, delivery_date`	The PO to check. **No stock or sales figures are sent**: the server has them

Response

FIELD	MEANING
`decision_id`	Store it on the PO item. Needed for the depth bypass and the audit trail
`overall_verdict, verdict_code`	Worst verdict across the items (see the Functional Logic Guide for the rules)
`route_to`	`NONE` / `BUYER` / `CATHEAD` / `BMHEAD`. SAP's own workflow decides the actual person
`items[].max_qty`	Highest acceptable quantity, when the requested quantity is too high

FIELD	MEANING
`items[].reason_text, summary_text`	Plain-language reason to show in the work item
`data_delayed`	`true` if the real-time feed is behind the limit. SAP then continues as today

Timing

- The server answers within 1 second. SAP waits **at most 3 seconds**.
- No answer, an error, or `data_delayed = true` → SAP continues with the current process and marks the PO "not checked".

Depth check (built in SAP)

The depth check is **bypassed only when both** of these are true:

1. The quantity is one the tool approved: the reduced quantity (Max Permissible) for a **Single Concept** PO, or the recommended option quantity for a **Multi Concept** PO (option-quantity logic to be finalised).
2. That quantity is **below the minimum depth quantity** in SAP for the option.

In every other case the normal depth check applies. If it fails, the usual depth change process runs: the category raises a depth change request to the planning team, and the planning team decides whether to allow it based on the business case.

CASE	DEPTH CHECK
Tool-approved qty, below min depth	**Bypassed**; PO continues
Tool-approved qty, at or above min depth	Normal check (it will pass)
Any other qty that fails depth	Usual process: category requests a depth change → planning decides

How SAP applies the bypass

1. When the tool returns a lower quantity, SAP stores `decision_id` and `max_qty` (or the option qty) on the PO item.

2. On resubmission, SAP skips the depth check for that item only if the new quantity is **at or below the tool's quantity**, **below the minimum depth**, and the item still carries that decision ID (same PO, item and day).
3. SAP resubmits to the tool with `depth_check = BYPASSED` and `previous_decision_id`. Every bypass is logged in SAP and in the tool.

07 How the server calculates

Stock on the server

```
Live stock = Yesterday's closing stock (BW)
  + GRC posted today (live from R/3)
  − GRT posted today (live from R/3)
```

Pending PO on the server (released POs only)

```
Pending PO = Previous day's released pending PO qty (BW, 04:00)
  + POs released today (live from R/3)
  − POs un-released today: release revoked (live from R/3)
  ± Qty changes on released POs today (live from R/3)
  − Released POs closed / deleted today (live from R/3)
  − GRC posted today against released POs
  + GRT posted today against released POs that are still open
```

WHY GRC IS DEDUCTED FROM PENDING PO

A GRC moves quantity from "on order" to "in stock". Example: a released PO of 1,000 pcs; 600 are received today. Stock goes up by 600, so pending PO must go down by 600 (to 400). If it does not, the same 600 pcs are counted twice, once in stock and once in pending PO, and cover is overstated.

Then, per article-colour

```
Pending PO is split by delivery month (from the PO delivery date)
Colour demand per month = Plan × colour's share of its MC-MRP sales (last
45 days)
Then the same month-by-month roll as v89: stock + pending PO − demand, up
to Day 1 of the month after delivery, then cover = stock ÷ daily rate
Next morning at 04:00: the new BW stock and released-PO baseline replace
everything; the previous day's adjustments are dropped
```

Reading the live position for a decision

1. Check the feed delay. If it is above the limit (e.g. 60 seconds), return `data_delayed = true`.
2. Read stock and released pending PO for each item's article-colour in **one consistent read**.

3. **Leave out the PO being checked.** SAP saves the PO before the workflow runs, so its own message may already be in the database.
4. Run the v89 rules (Single Concept or Multi Concept).
5. Write the decision log: inputs used, last message `seq` applied, rules version, verdict. Then reply.

Daily timeline (example: today is 9 Oct)

TIME	BW	SAP R/3 LIVE FEED	SERVER
9 Oct 00:00	—	9 Oct business day starts. Every PO / GRC / GRT is sent as it happens	Still on the 7 Oct snapshot + 8 Oct messages; now also adds 9 Oct messages
00:00 – 04:00	—	Messages continue	Applies them and stores them for replay
04:00	Job runs: extracts the **8 Oct closing** stock, released pending POs (receipts / returns dated up to 8 Oct), sales and masters	Messages continue	Waits for the control file
After files arrive	Control file written	—	Switches in one step: load the 8 Oct snapshot, drop all 8 Oct messages, re-apply all 9 Oct messages (from 00:00)
Rest of 9 Oct	—	Messages continue till 24:00	8 Oct snapshot + 9 Oct messages
10 Oct 04:00	Extracts the 9 Oct closing	—	Same switch: 9 Oct messages dropped, 10 Oct messages kept

No double counting: which messages are applied

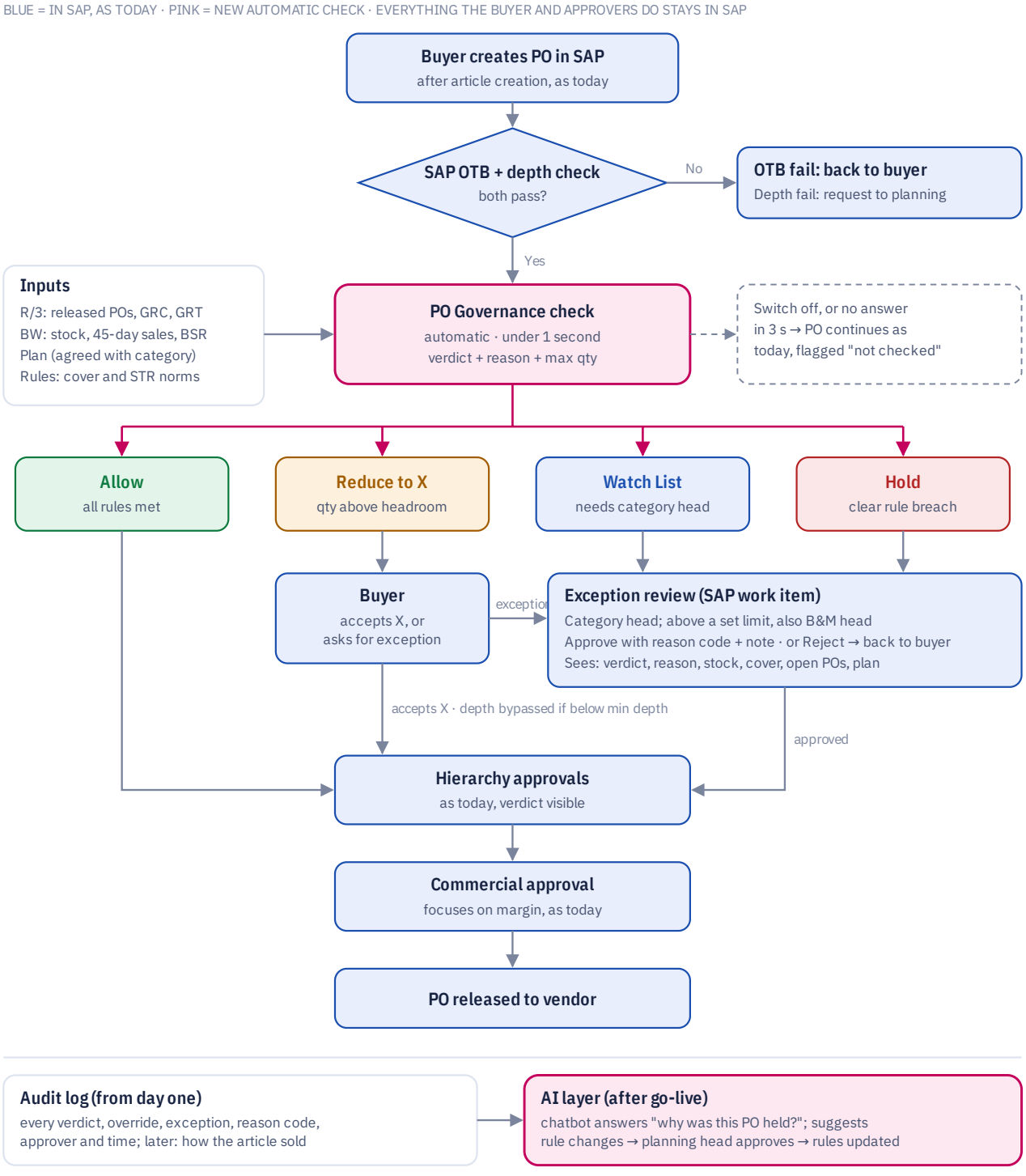
The business day runs 00:00 to 24:00. After loading the snapshot of day D−1 (e.g. 8 Oct):

MESSAGE	APPLIED?	WHY
GRC / GRT with posting date = today (9 Oct)	**Yes**	Not in the 8 Oct snapshot
GRC / GRT with posting date ≤ 8 Oct, saved in SAP before the 04:00 extraction	No	Already in the 8 Oct snapshot
GRC / GRT with posting date ≤ 8 Oct, saved after 04:00 (back-dated entry)	**Yes**	BW had already run, so it missed it. Next day's snapshot includes it and the adjustment is dropped then
PO messages (create / change / release / close)	**Yes**, always	A PO message carries the whole PO and simply overwrites ordered qty, delivery date and release status. Applying it twice gives the same result

```
Pending qty per released PO line = Ordered qty (latest PO message, else
baseline)
  − Received qty (baseline up to 8 Oct + GRC applied today)
  + Returned qty (baseline up to 8 Oct + GRT applied today)
Only lines whose latest status is released, not closed, not deleted are
counted
```

To confirm with the BW / SAP team: at what time the previous day's store sales (POS) are fully posted in SAP. The 04:00 BW job must start only after that, otherwise the 8 Oct stock and sales will be incomplete.

08 PO workflow in SAP



#	STEP	WHERE	WHAT HAPPENS
1	Buyer creates the PO	SAP	As today
2	OTB check	SAP	As today. Fails → back to buyer

#	STEP	WHERE	WHAT HAPPENS
3	Depth plan check	SAP	As today. Fails → category requests a depth change, planning decides on the business case
4	PO Governance check	Tool	Verdict with reason and max qty
5	Allow	SAP	Continue to hierarchy approvals
6	Quantity too high	SAP	Back to buyer with max qty. Buyer accepts it (depth check bypassed if below min depth) or asks for an exception
7	Watch List / Hold / Block	SAP	Exception to the category head (B&M head above the limit): approve with reason code, or reject
8	Hierarchy approvals, then commercial (margin), then release	SAP	As today. Once released, the PO counts in open PO

09 Stage 3: AI layer

Built after Stages 1 and 2 have run steadily. It reads data the server already stores. No new SAP interface.

FUNCTION	WHAT IT DOES	DATA IT READS
Decision assistant	Answers questions like "why was this PO held?" and shows the record behind the answer	Decision log, rule book, live position
Exception review	Monthly: groups exceptions by reason code, category, buyer, vendor, and checks how those articles sold after 30 / 60 / 90 days	Decision log, `GOV_OUTCOME` messages, BW sales and stock
Rule suggestions	Proposes rule changes with evidence. The planning head approves or rejects. Approved changes become a new, dated rule version	Exception review results, rule book
Daily insights	Short morning summary: overstock risks, large pending POs, slow sellers with open POs	Live position, sales

FUNCTION	WHAT IT DOES	DATA IT READS
Pattern alerts	Flags unusual behaviour, e.g. one requirement split into several POs, repeated exceptions by one buyer or vendor	Decision log, PO history

Limits: the AI is read-only. It never approves, blocks or changes a PO, and never changes a rule without approval. Every suggestion is logged.

10 Data maintained in the tool

Everything else comes from BW or SAP (section 04). Only these are set or uploaded in the tool:

DATA	HOW IT GETS THERE	USED FOR
Plan	Manual upload by planning now; later fetched hourly from SAP SFCC / SFCR	Future demand per MC-MRP-Flag-month
Attribute Relevance Master	Manual upload by planning	Multi Concept recommendations
SCR mapping / category heads	Manual upload	Logins for buyers and category heads; approval routing
Parameters	Set by planning in the tool	Thresholds (cover limit, buffer, window, mix gate, WOF gate, peer minimum, border zone)
Season selection	Chosen by the user. Current season(s) can be combined (e.g. AW26 + SS26). Fallback = same season last year (AW26 → AW25, SS26 → SS25)	BSR peer pool
Decision log and exception reasons	Written by the server	Audit trail and Stage 3

The manual upload screens stay available for every BW / SAP file as a fallback.

11 If the server is down

SAP runs the current process automatically. Nobody has to do anything.

LAYER	WHAT HAPPENS
Per PO	No answer in 3 seconds, an error, or `data_delayed` → PO continues as today, marked "not checked"
Automatic switch-off	An SAP job calls `/ping` every minute. After 3 failures in a row it switches the governance step off for all divisions and alerts IT and planning
Manual switch	Parameter `ZPOGOV_ACTIVE` per division: blank = off, `S` = shadow, `L` = live. Changed in seconds, no transport
No messages lost	bgRFC keeps every message while the server is down and delivers them in order afterwards
Switching back on	Manual, after `/ping` is healthy and the queue has caught up
Server	Runs as a Windows service that restarts itself; database backed up daily; standby server optional
BW files fail	Previous base plus live messages is kept (last weekly BSR stays in use); manual upload available

12 API summary

ENDPOINT	CALLED BY	PURPOSE
`GET /ping`	SAP Basis test; SAP health job	Connection test and automatic switch-off signal
`POST /events`	SAP bgRFC queue	PO, GRC, GRT and `GOV_OUTCOME` messages
`POST /decisions`	SAP workflow step	Verdict for a submitted PO
`GET /reason-codes`	SAP approver screen	Exception reason codes

- **Base URL:** `https://<server>:8443/api/sap/v1`, plus a separate test server for SAP QA.
- **Security:** HTTPS with the company certificate; Basic authentication with one technical account per SAP system; IP allow-list of the SAP application servers. SAP always calls the server; the server never logs in to SAP.
- **Format:** JSON, UTF-8, snake_case field names, ISO 8601 timestamps, quantities in base unit.
- Full field list, examples, error codes and ABAP sketches: **PO_Governance_SAP_API_Specification_v1.4.pdf**.

13 Terms

TERM	MEANING
GM · article · colour	Generic Material (style), variant article, colour
MC · MRP · Flag	Merchandise category, price point, BOGO / non-BOGO
Single Concept / Multi Concept	Single-style MRP (judged on stock cover) / multi-style MRP (judged on peer ranking)
Cover (STR)	Days the stock will last at the planned daily rate
Max qty (Max Permissible)	Highest quantity that keeps cover within the limit
Open PO (released pending qty)	Pending quantity of released, not closed vendor POs
Min depth qty	The minimum quantity per option in SAP's depth plan
BSR	Weekly sell-thru snapshot from BW
GRC / GRT	Goods receipt from vendor / goods return to vendor
Snapshot date / extraction time	The business day BW's data closes (e.g. 8 Oct) / when the BW job actually ran (9 Oct 04:00)
Business day	00:00 to 24:00; the live feed covers the current business day
bgRFC	SAP's ordered, retrying outbound queue

TERM	MEANING
Shadow mode	The tool decides and records; SAP does not act on it yet